

Ex. No 5 Use Autopsy to create a case and import evidence

DATE:

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices.

Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link: https://drive.google.com/drive/u/1/folders/1lSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Download files: [4Dell Latitude CPi.E01](#), [4Dell Latitude CPi.E02](#)

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.

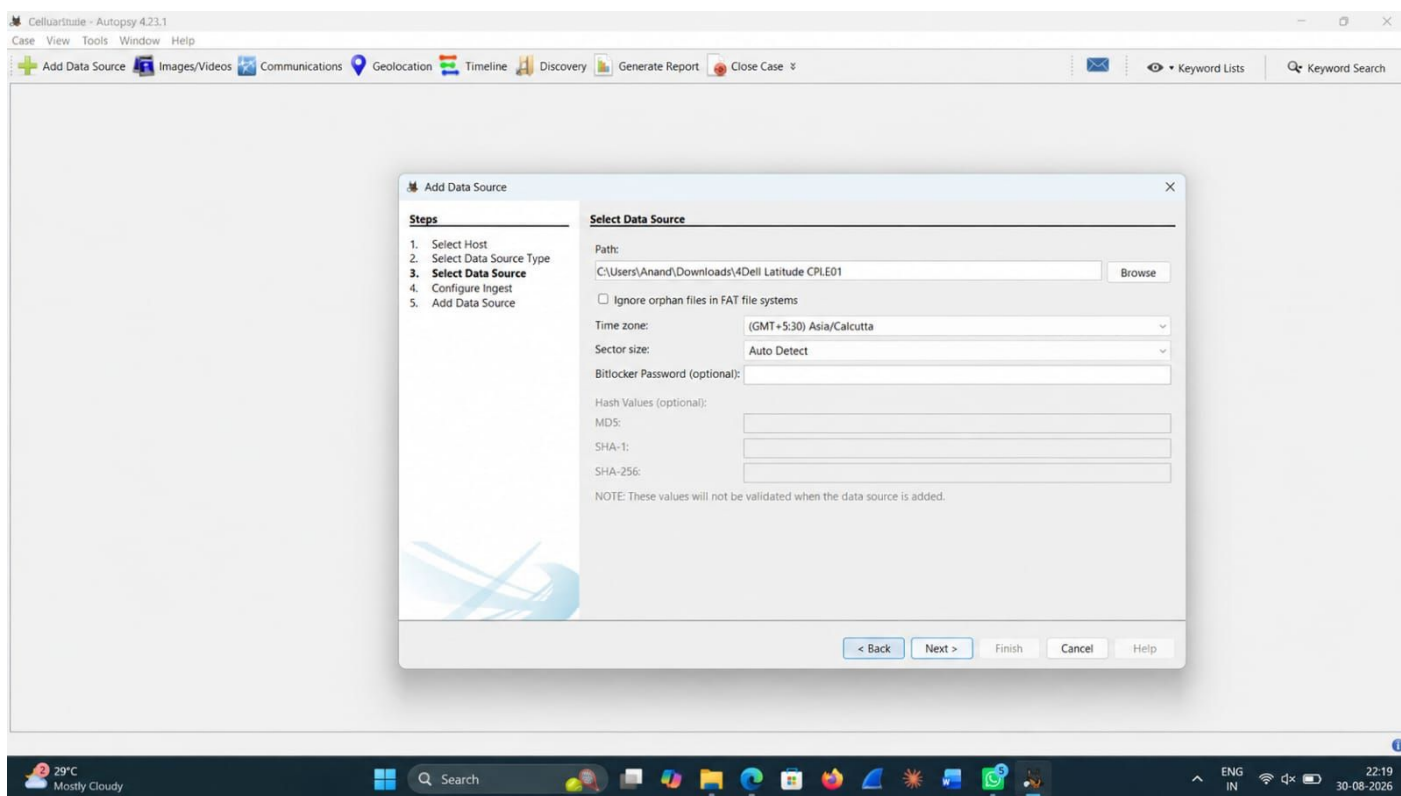
The screenshot shows the 'New Case Information' window in the Autopsy application. On the left, a 'Steps' sidebar lists '1. Case Information' and '2. Optional Information'. The main area is titled 'Case Information' and contains the following fields and options:

- Case Name:** A text box containing 'delllaptiude'.
- Base Directory:** A text box containing 'C:\Users\Anand\OneDrive\Desktop'. To its right is a 'Browse' button.
- Case Type:** Two radio buttons: 'Single-User' (selected) and 'Multi-User'.
- Case data will be stored in the following directory:** A text box containing 'C:\Users\Anand\OneDrive\Desktop\delllaptiude'.

At the bottom of the window, a red error message reads: 'ERROR: The Base Directory that you entered is not valid. Please enter a valid Base Directory.' Navigation buttons at the very bottom include '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

3. Adding a Data Source

- **Choose the Type of Data Source:**
 - After creating a case, you'll be prompted to add a data source.
 - You can add disk images, directories, logical files, or local disks.
- **Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- **Configure Ingest Modules:**
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.



4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.

- **Use the Tree Viewer:**

- On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**

- You can perform specific keyword searches using the Keyword Search module.
- Use pre-configured lists or enter custom keywords.

- **File Analysis:**

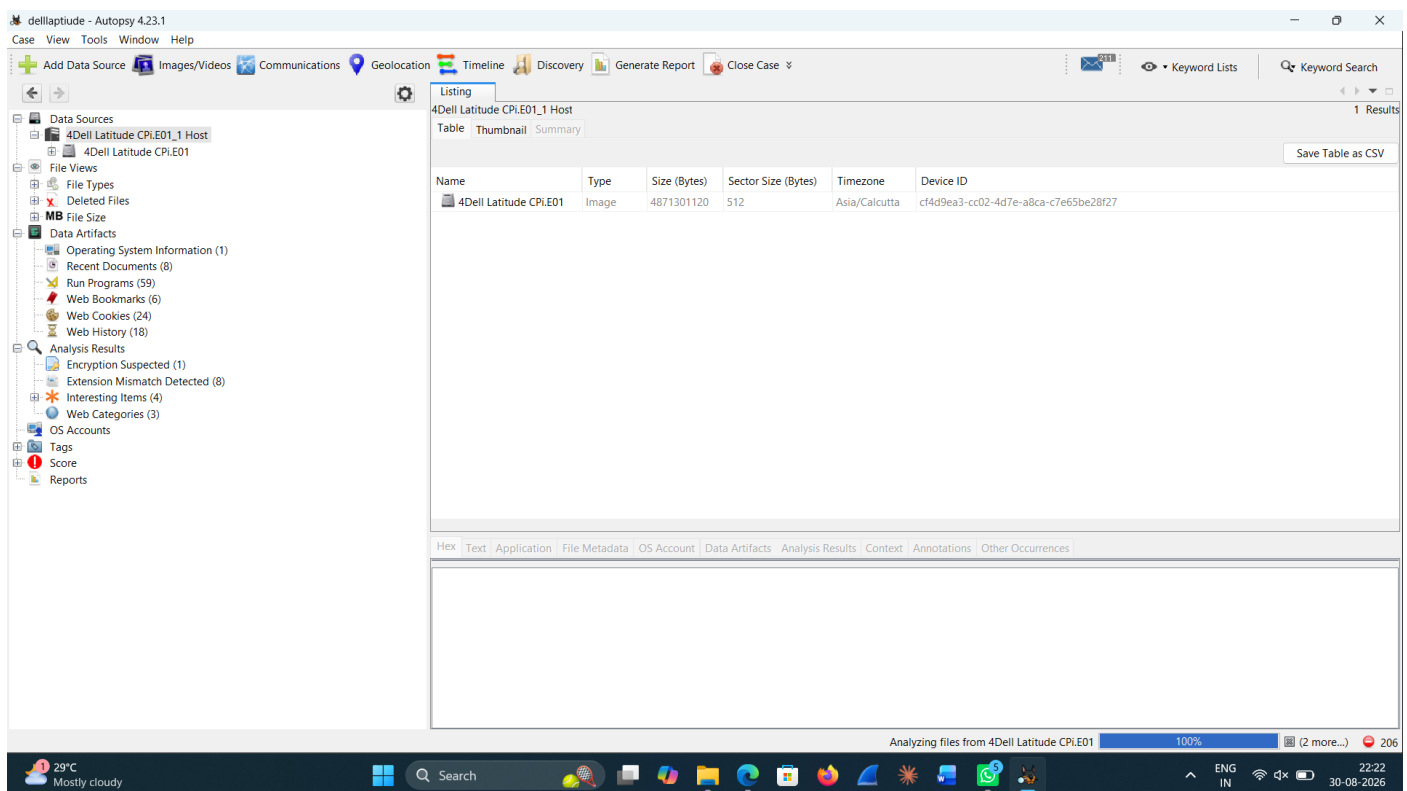
- Navigate through files and folders under the File Types or File System section.
- Open, view, or export files for further examination.

- **Timeline Analysis:**

- Use the Timeline module to visualize events based on timestamps.
- This can help track user activity over time.

- **Hash Analysis:**

- Compare file hashes with known databases to identify known good or bad files.



6. Reporting

- **Generate a Report:**

- After analyzing the data, click on Generate Report from the toolbar.

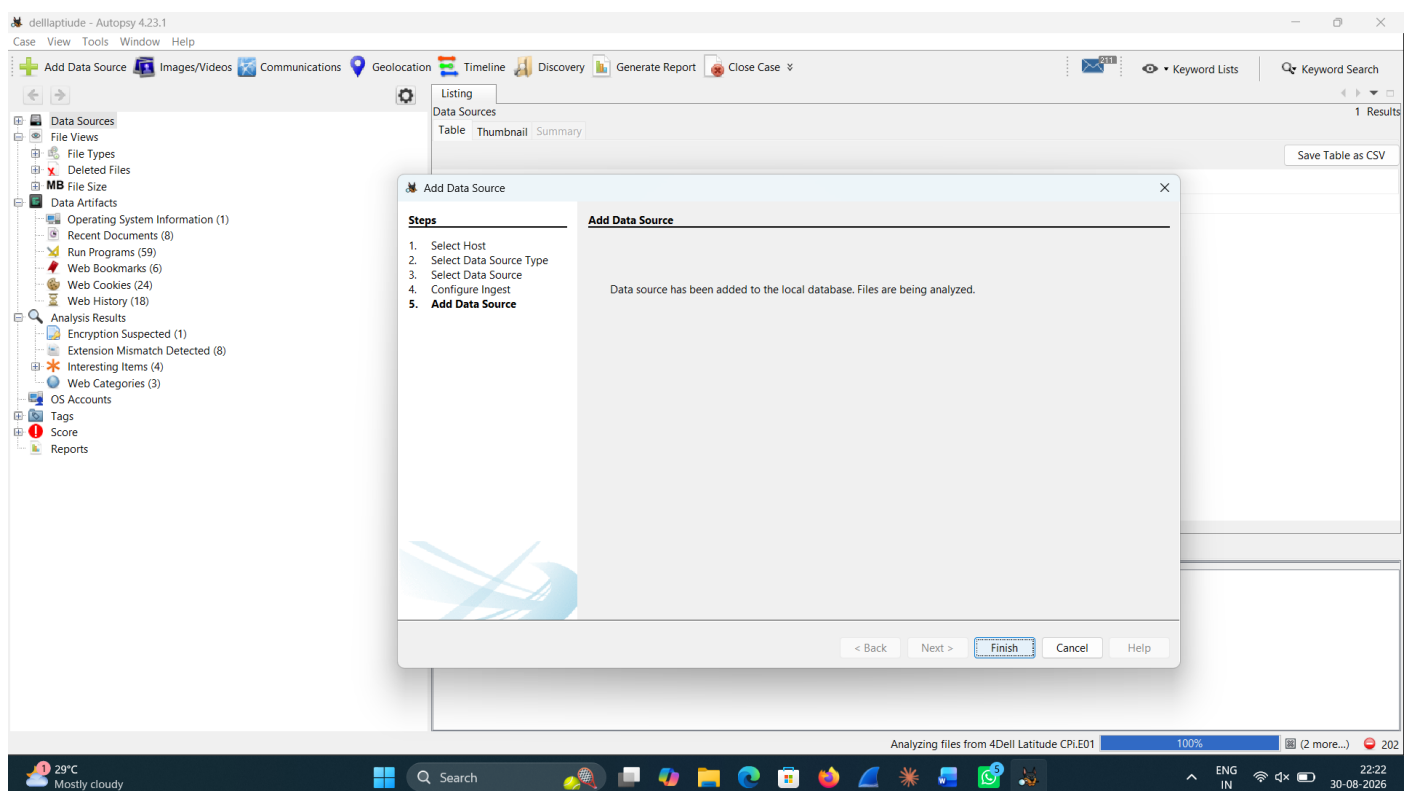
- Choose the type of report (HTML, CSV, Excel, etc.).
- Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Successfully Completed. The forensic case was created in Autopsy, the disk image was ingested and analyzed, deleted credentials were recovered, and a comprehensive Excel report was generated.